Infra-Red Radiation



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Infra-Red Radiation (IRR)

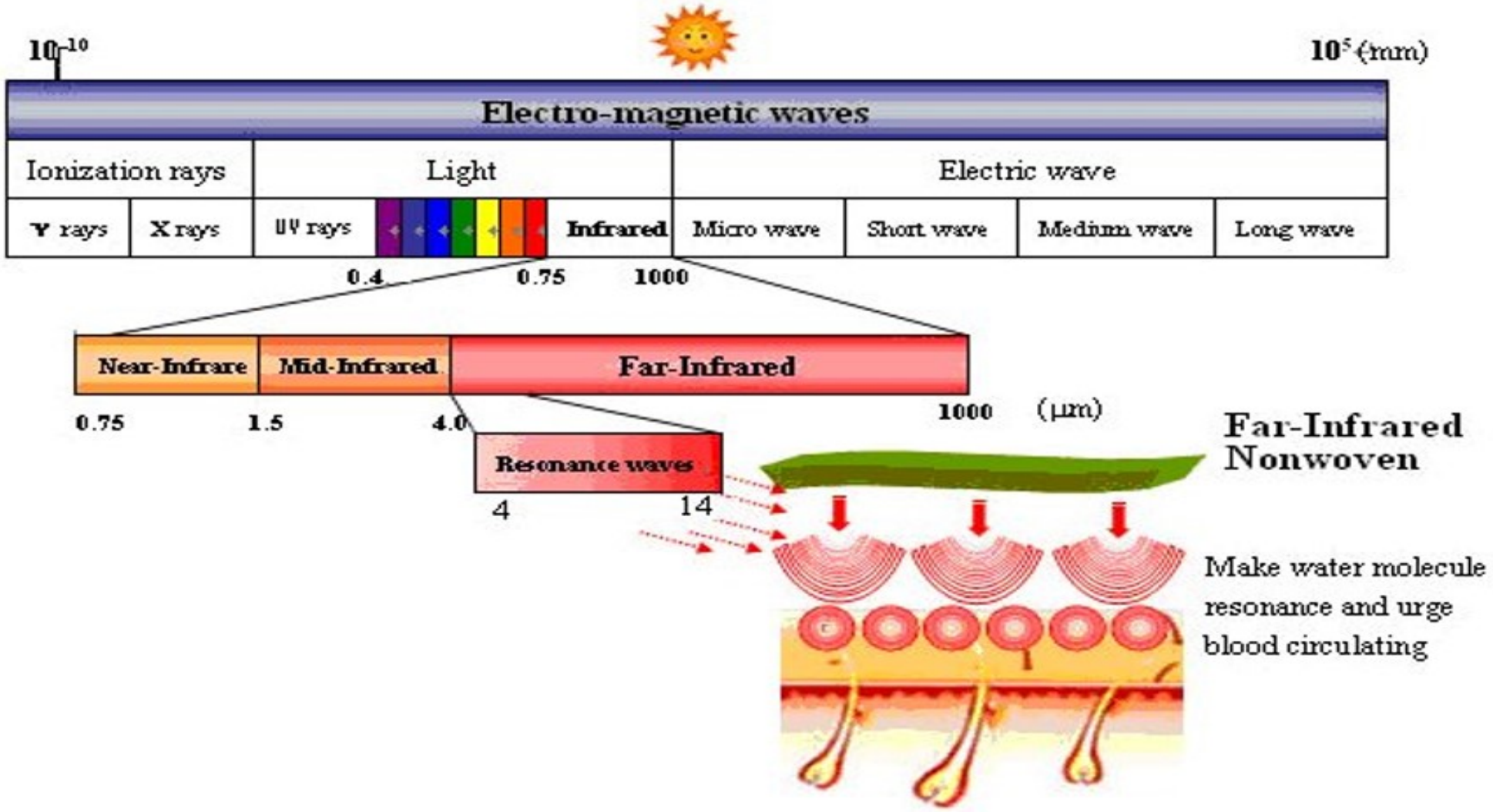
Definition:

*Infra red rays are electromagnetic wave with wavelengths 760:15000 nm (nanometers). It lies between microwave and visible light in the electromagnetic spectrum.

*Any heated body emits infrared radiations that can be subdivided into three region or bands, A, B, C according to its wave length.

Classification of infrared radiations

Type	Wave length
IRA	760-1400 nm
IRB	1400-3000 nm
IRC	3000-15000 nm (not used in therapy
Former classification	
near or short IR	760-1500 nm
far or long IR	1500-15000 nm



Source: http://www.quantumacceleration.net/BioMat.html

Production of infrared radiations

Infrared radiations are produced as a result of molecular motion within material. An increase in temperature above absolute zero results in the vibration or rotation of molecules within the matter, which lead to the emission of infrared radiations.

Sources of infrared radiations

Luminous generators

emit visible light, IRR and few ultraviolet (UV).

Luminous lamps are produced by electrically heated filament in an evacuated glass bulb or may contain an inert gas at a low pressure, often with silvered inner surface to provide a reflector

Non-luminous generators

provide I.R. rays only

produced by electrically heated resistance wire coiled on a cylinder of insulating material such as a fireclay or porcelain. The resistance wire serves as the heater, and the cylinder becomes the radiation source.

Luminous generators

The luminous lamps emitted approximately; 70% short IR, 24% long IR, 5% visible light and a few 1% UV rays.

Often the front of the bulb is red to filter out the UV rays.

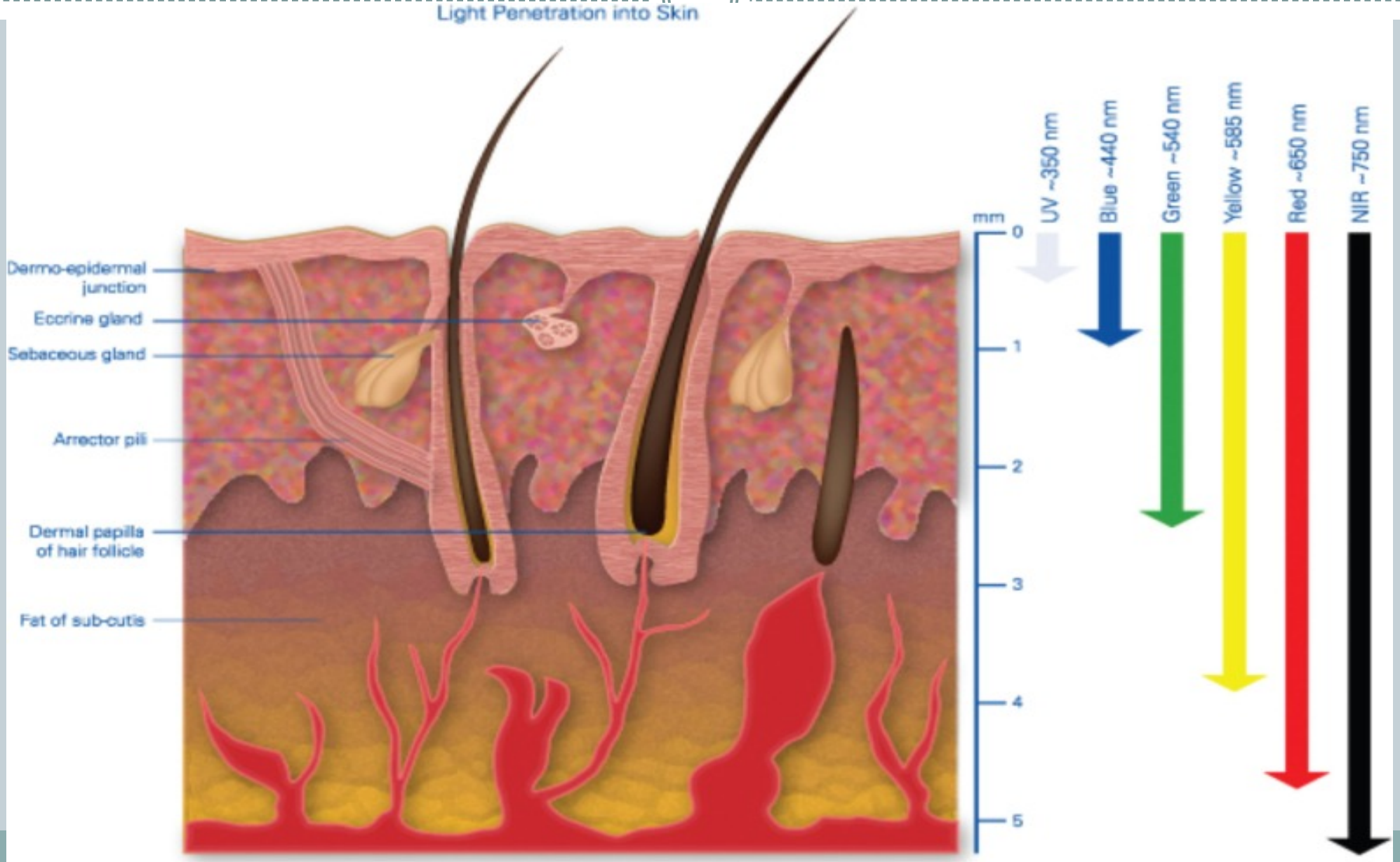
Non-luminous I.R generators :

These lamps emitted mainly long IR radiation around 3000-4000 nm with about 10% short IR.

1. The elements of the non-luminous lamps take about 5: 10 mins to heat up and emit their maximum intensity, so lamps must be switched on at an appropriate time before they are required.

2. In all radiators there are always some visible light rays emitted as a red glow, it is visible when the element is hot. So they are not completely non-luminous.

3. It has been suggested that the non-luminous units apparently feel hotter than do the luminous at equal distances and power levels, due to the increased absorption of the longer wave lengths by the top layers of the skin.


Light Penetration into Skin
Dermo-epidermal junction
Eccrine gland
Sebaceous gland
Arrector pili
Dermal papilla of hair follicle
Fat of sub-cutis
mm
0
1
2
3
4
5
UV ~350 nm
Blue ~440 nm
Green ~540 nm
Yellow ~585 nm
Red ~650 nm
NIR ~750 nm

- Red light penetrates deeper than blue light, The reason is that skin consists of a range of chromophores which have scattering and absorption coefficients which are highly wavelength dependent.
- The scattering properties of tissue are due to attenuation properties intrinsic to the chromophore and also to the size of the particles within the tissue which also governs the type of scattering that occurs, namely Mie or Rayleigh scattering

- Scattering leads to light distribution in the tissue and the eventual reduction in the energy density with increasing depth

Tunnel Bath (electric light bath) = Bakers:

There is upright tube that consists of a metal or wooden cabinet fitted with a number of incandescent lamps arranged among the inner sides

Used for chronic conditions.

Several areas can be treated.

Tunnel Bath



Absorption and penetration of IRR:

Absorption will take place within the top 3mm of tissues, and the heat is carried to the deeper tissues by conduction and by the circulating fluids.

The absorption of IRR and the maximal penetration of the rays will depend upon the following variables:

Variables affect I.R absorption

1-Frequency or wave length of the rays (short IR can be penetrated to the level of dermis and subcutaneous tissues while long IR penetrates only to the epidermis level). .
2-Thermal conductivity of the tissues
3-Density of each tissue.

4- Specific heat of each tissue.

5- angle of incidence of the rays.

6- Distance from the source of I.R.

7- Patency of the circulation.

8- Source of the I.R.

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Physiological effect of IRR.:



1- Nerve stimulation:

IRR stimulates thermal heat receptors in the skin which gives awareness of the heating.

Increase sensory nerve conduction velocity which influences sensory response via increase in endorphin release.

2- Blood supply:

IRR increase blood supply in the superficial tissue. This effect is due to vasodilatation of capillaries caused by increase liberation of histamine-like substances.

The heat regulating center in medulla also signals the capillaries to vasodilator when the temperature rises).

- ***3- Phagocytosis:*** this process increased with temperature, and if there is superficial inflammation, heating will promote phagocytosis.

4- Pigmentation and erythema:

- IRR causes a reddening of the skin, which is a gentle erythema that disappears rapidly. This erythema is due to distraction of red blood corpuscles. If radiations are given frequently, the skin pigments in a mottled fashion.

5- Reflex heating:

IRR can be applied to the abdominal area to promote peripheral circulation (heating of the large splanchnic vessels in the abdomen that stimulates the heat regulating center in the medulla which reflexively opens up the peripheral vessels in an endeavor to regulate the body temperature quickly.

6- Sweating: with prolonged or intense heating the activity of sweat glands increased by reflex stimulation from the heat regulating centre to cool the skin.

7-Blood pressure: if heating is given to a large area of the body for a prolonged period as in the use of IR baker, there will be a fall in blood pressure due to the generalized vasodilatation, and reduction of peripheral resistance in the arterioles.

8-Increase efficiency of muscle action: rise in temperature causes muscle fibers to contract and relax more quickly.

Metabolism: metabolic rate increase. Heat produced by IR increase the demands to oxygen and food stuff as well as increases output of waste products.

Therapeutic effects and indication of IRR:

1-Relief of pain:

Pain may be due to accumulation of waste product of metabolism in the tissue so increased blood flow through the part by IR lead to removal of chemical mediators which causes irritation of nerve ending and lead to relief of pain.

Sedative effect: mild heating increases skin temperature that lead to state of analgesia by increase pain threshold.

Counterirritant effect: Strong heating stimulates the superficial sensory nerve fibers which blocking the transmission of pain.

Therapeutic effects and indication of IRR:

2-Reduce of muscle spasm: ??????????

3-Relaxation of Muscle: muscle relaxes when tissues are warm and also relief of pain facilitates relaxation. So IR can be used prior to other forms of physical therapy.

Therapeutic effects and indication of IRR:

4- increase range of motion and decrease joint stiffness:

IR increases extensibility of soft tissue that occurs with increasing its temperature.

It's preferable to perform stretching exercises immediately after application of IR.

5- Edema:

IR reduces chronic edema of the hand and foot, but with elevation during treatment. Vessels dilation induced by heating will allow increased rates of fluid exchange and thus may help to increase the reabsorption of exudates

Therapeutic effects and indication of IRR:

6-Healing of tissue:

IR accelerates tissue healing by increasing circulation and enzymatic activity. A good blood supply to the tissue is essential for healing to take place. Increase blood supply brings more oxygen and nutrients to tissue and removes waste products.

Increasing the temperature of blood by IR increases the dissociation of oxygen from hemoglobin to tissue, making more oxygen available for the process of tissue repair.

If there is an infection the increased number of white blood cells (WBC) and the phagosetic effect of IR assist in destroying the bacteria.

7-treatment of arthritic joints and other sub acute or chronic inflammatory lesions

Contraindications to IRR:

- ***Impaired sensation:*** patients with impaired sensation in the area to be treated will not be able to determine if excessive heating is occurring. Large areas of scar tissue with impaired sensation will also be a contraindication.
- ***Acute injury or acute inflammation:*** as increasing tissue temperature can increase edema and bleeding as a result of vasodilatation and increased blood flow. This may aggravate the injury, increase pain and or delay recovery.
- ***Acute infection: all*** acute infections are a contraindication to IR, as the increase in temperature is likely to exacerbate the infective process.

Contraindications to IRR:

4- Impaired circulation (defective arterial blood supply): when there's a history of defective circulation from any circulatory disease (e.g. atherosclerosis, deep venous thrombosis (DVT) or Burger's disease), care must be taken not to administer heat over the area with impaired circulation.

The function of skin circulation is to dissipate heat, and if this heat regulating mechanism is defective, it would be quite easy to cause a burn.

5- Dermatological conditions: skin lesions such as fungus, dermatitis and eczema are some lesions to look for. Heat tends to irritate skin lesions.

Contraindications to IRR:

6- Metal: there should be no metal in the area that is irradiated. Metal retains the heat and will cause a burn to the underlying tissue.

deep metal implants are not a contraindication.

7- Eyes: radiations can cause cataracts. This will occur if I.R. R is given over a long period. It can also cause iritis.

8- Topical creams and oils (recent use of ointment): all applications must be removed before giving I.R.R, or the creams and oils will cause burns because they make the skin hypersensitive.

9- Skin tumors: patients with skin tumors or melanoma must not receive I.R.R, as the tumor growth may be increased.

Contraindications to IRR:

10- Blood pressure abnormalities: I.R.R should not be given to large areas for a prolonged time, as the patient may be unable to tolerate the change in BP which may be produced.

11- Sever cardiac conditions: heating a large area will cause an increase in cardiac output which may not be tolerated by patients with sever cardiac conditions.

12- Unreliable patients: - Very young people: unreliable in reporting the heat intensity, there is a tendency to fall asleep, and they lack the normal C.V and respiratory reserves.

Contraindications to IRR:

13- Very old people: they generally have impairment of sensation and circulation, and in these patients large areas must not be irradiated.

14-The testes: temporary lowering of sperm count.

15-Analgesic and narcotic drugs: if the patient have had strong analgesic or narcotic drugs just prior to treatment, I.R.R must not be given as these drugs will raise the pain threshold and the patient will not be able to determine whether the I.R.R are of too great an intensity.

16-Deep X-Ray therapy: for 3 months after therapy, patients must not be given I.R.R, as deep X-Ray therapy reduces sensory appreciation.

17-Danger of hemorrhage (hemophilia).


THANK YOU